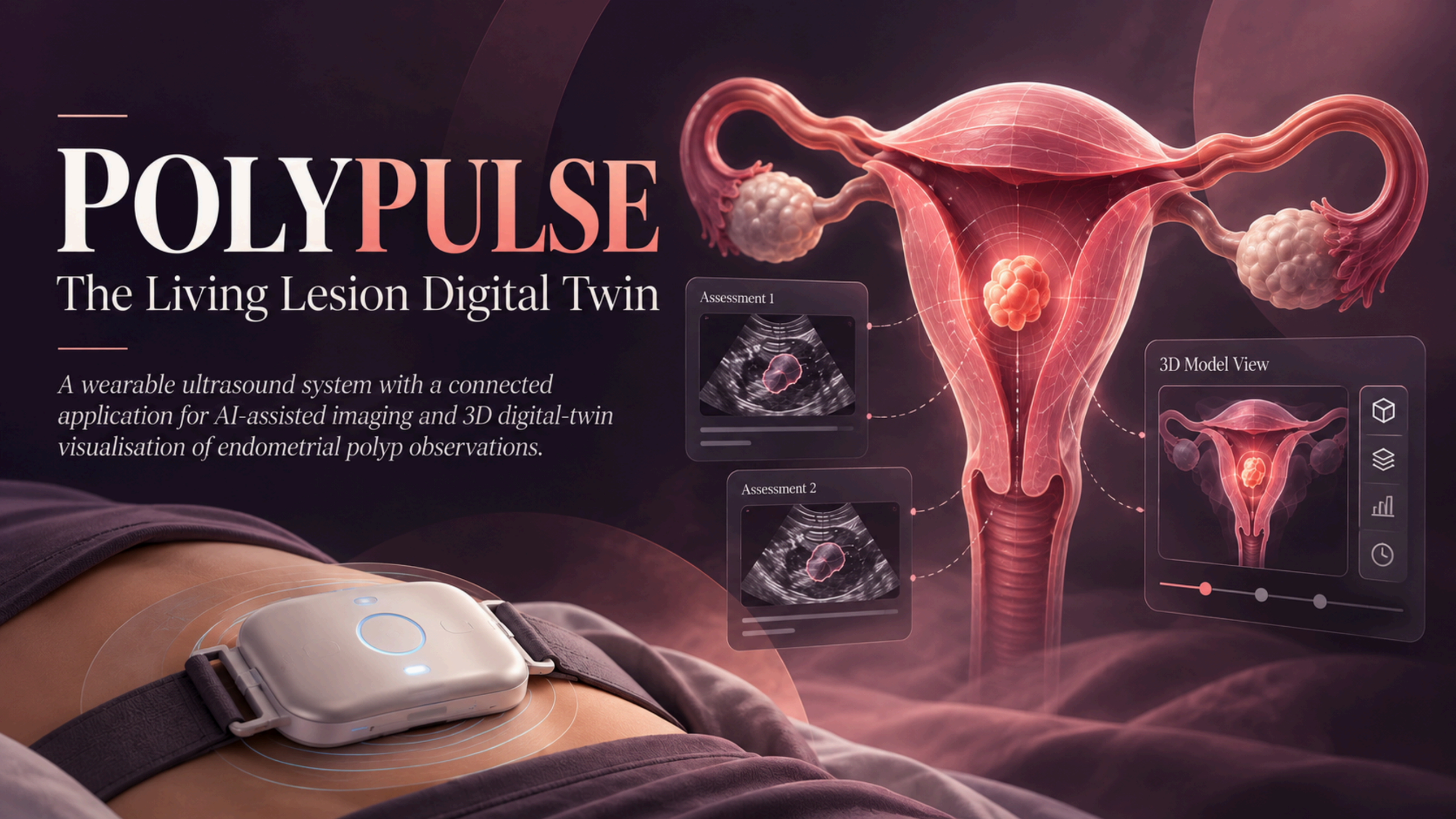
POLYPULSE

The Living Lesion Digital Twin

A wearable ultrasound system with a connected application for AI-assisted imaging and 3D digital-twin visualisation of endometrial polyp observations.



PROBLEM STATEMENT

The Challenge of Longitudinal Follow-up

Endometrial polyps often require repeated monitoring over time, but current follow-up is fragmented, inconsistent and difficult to compare.

Fragmented Information

Ultrasound images, measurements and reports are collected at different times and stored separately.

Difficult Comparison

Changes in polyp size, shape and morphology across multiple assessments can be challenging to track and compare.

Need for Better Organisation

A structured and visual approach can support more informed longitudinal review and clinical decision-making.

Our Solution POLYPULSE

A wearable ultrasound system with a connected application for 3D visualisation and longitudinal tracking of endometrial polyp observations.

1 Wearable Ultrasound Device

Captures ultrasound images through a portable, wearable system.

→

2 Connected Application

Transfers, stores and organises imaging data in a unified interface.

→

3 AI-assisted Analysis & 3D Reconstruction

Analyses images and reconstructs a 3D model of the uterine anatomy.

→

4 Longitudinal Tracking

Compares observations over time to track changes in polyp size and morphology.

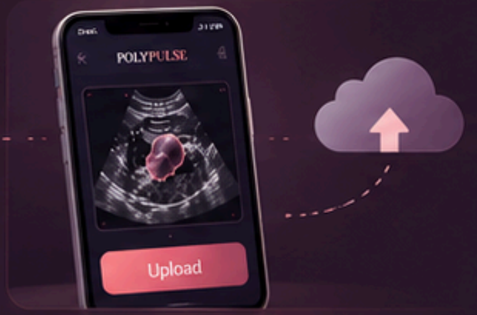
How It Works
From Scan to Insight

A simple and connected workflow to capture, visualise and track endometrial polyp observations over time.

1 Capture
Wearable Ultrasound Device

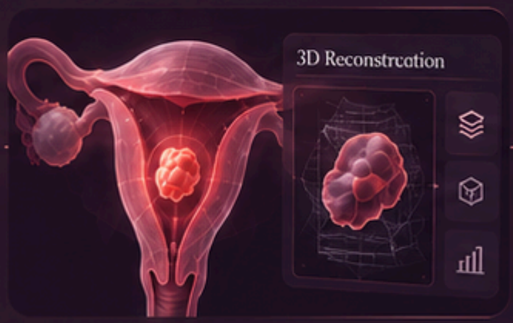
Captures ultrasound images of the uterus at regular intervals through a portable, wearable system.

2 Transfer
Connected Application



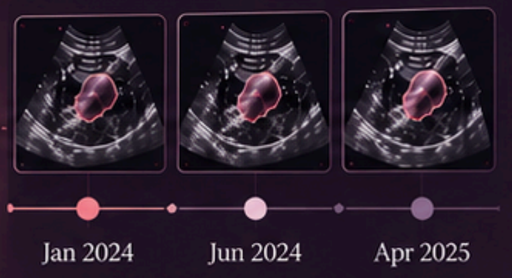
Images are securely transferred, stored and organised in a unified application.

3 Analyse & Reconstruct
AI-assisted Processing



AI algorithms analyse the images and reconstruct a 3D model of the uteeine anatomy with the polyp.

4 Track & Compare
Longitudinal Insights



Observations are compared over time to track changes in polyp size and morphology.

System Architecture & Technology Stack

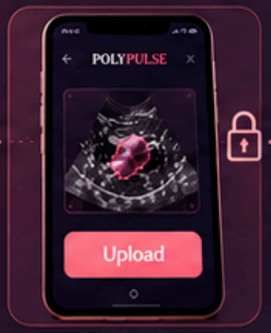
An integrated system that connects wearable ultrasound, AI analysis and a unified application for longitudinal tracking of endometrial polyps.

System Architecture

1. Data Capture
Wearable Ultrasound Device

Captures ultrasound images at regular intervals.

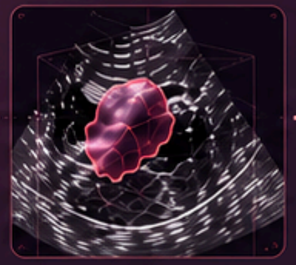
2. Data Transfer
Secure & Encrypted



Transfers data to a unified application.

Cloud Storage & Data Management

3. AI Processing
Analysis & 3D Reconstruction

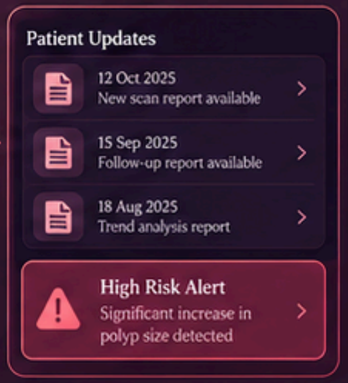


AI algorithms analyse images and reconstruct a 3D model.

4. Insights & Tracking
Web / Mobile Dashboard

Visualises, tracks and compares changes over time.

5. Doctor Dashboard
Reports & Alerts



Time-to-time reports are automatically sent to the doctor, with alerts in case of high risk.

Technology Stack

Frontend	React, Vite, TypeScript, Tailwind CSS
Backend	FastAPI, Python
AI / Image Processing	Python, OpenCV, 3D Reconstruction Libraries
Database & Storage	PostgreSQL, Cloud Storage
Deployment	Vercel / AWS / Cloud Platform

Hardware Technology Stack

A proposed wearable ultrasound system designed to capture high-quality uterine images for seamless integration with the POLYPULSE application.

Ultrasound Imaging

High-frequency transducer to capture uterine images (selection in progress).

Embedded Processing

On-device data acquisition and preliminary processing (microcontroller/SoC under evaluation).

Communication

USB / Bluetooth / Wi-Fi for secure data transfer to POLYPULSE application (final interface TBD).

Power Management

Rechargeable battery with charging and power regulation for safe and reliable operation.

Wearable Design

Compact, lightweight enclosure with comfortable body attachment and acoustic coupling.

Safety & Control

Temperature monitoring, electrical protection and ultrasound output controls to ensure safe operation.

End-to-End Flow

1. Capture
Wearable ultrasound device captures images at regular intervals.

→

2. Transfer
Images are securely transferred to the application.

→

3. Store
Data is organised and stored in a unified platform.

→

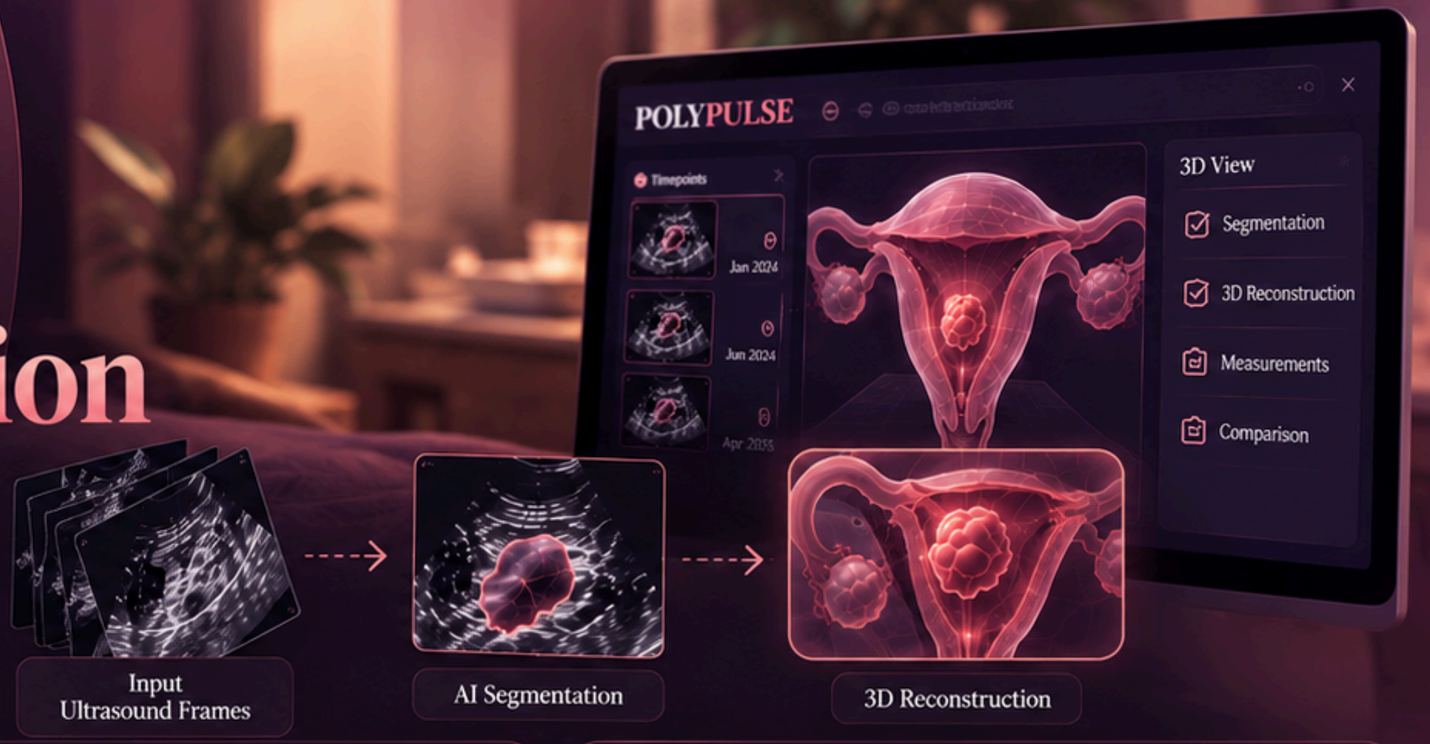
4. Analyse
AI reconstructs 3D model and tracks changes over time.

→

5. Visualise
3D digital twin and longitudinal insights in POLYPULSE.

AI Integration & 3D Reconstruction

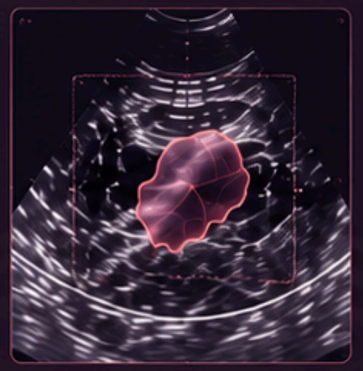
AI algorithms analyse ultrasound images to segment, reconstruct and track endometrial polyps over time, creating a 3D digital twin for longitudinal insights.



Input Ultrasound Frames → AI Segmentation → 3D Reconstruction

1 Autonomous Image Analysis

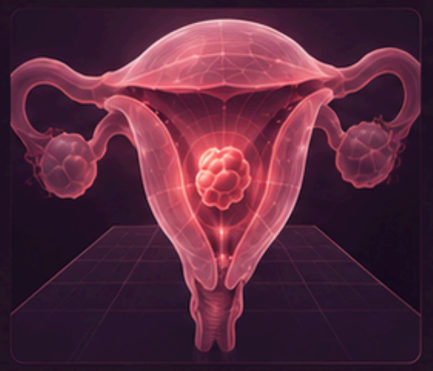
AI automatically processes ultrasound images to detect and segment endometrial polyps.



- Pre-processing (Noise Reduction)
- Polyp Segmentation (Deep Learning)
- Feature Extraction (Size, Shape, Morphology)

2 3D Reconstruction & Digital Twin

Segmented images are combined to reconstruct a 3D model of the uterus with the polyp.



- 3D Volume Reconstruction
- Morphological Analysis
- Quantitative Measurements

3 Longitudinal Tracking & Insights

3D models are compared over time to track changes in polyp size and morphology.

Visual comparison helps in longitudinal monitoring and objective assessment of changes over time.

System Workflow

From wearable ultrasound capture to a 3D digital twin for continuous monitoring of endometrial polyps.

1 Image Capture

Wearable ultrasound device captures uterine images at regular intervals.

2 Data Transfer

Images are securely transferred to the POLYPULSE application.

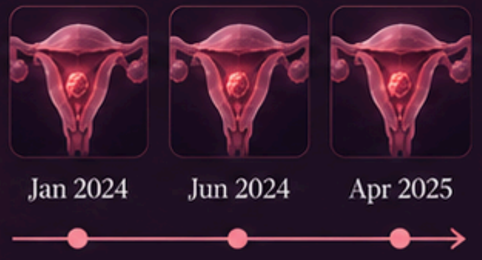
3 AI Processing

AI algorithms segment and analyse the polyp from ultrasound images.

4 3D Reconstruction

Segmented images are reconstructed into a 3D model of the uterus with the polyp.

5 Longitudinal Tracking



3D models are compared over time to track changes in polyp size and morphology.

Key Outcomes

Early Detection
Identify and monitor endometrial polyps over time.

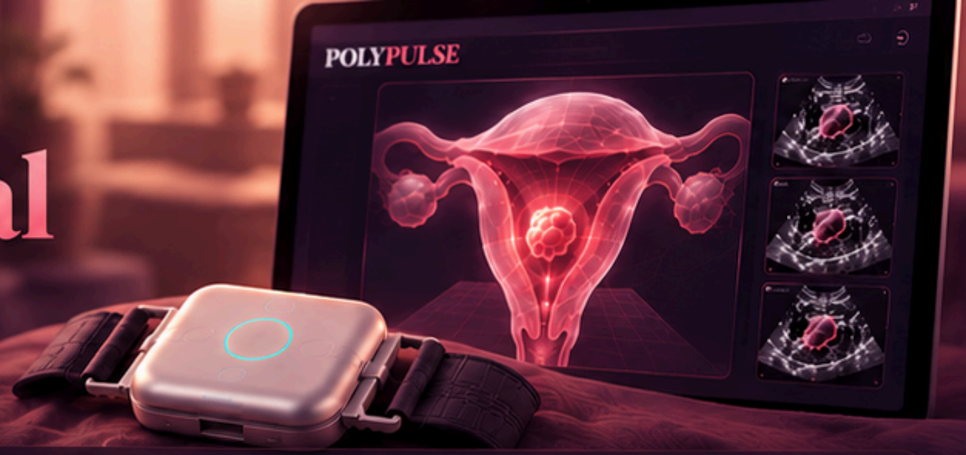
Objective Tracking
Quantitative comparison of polyp size and morphology.

Personalised Insights
3D digital twin for longitudinal analysis.

Improved Decision Support
Visual and quantitative data to support clinical evaluation.

Innovation & Future Potential

A novel, AI-enabled, wearable ultrasound platform with the potential to transform endometrial polyp monitoring and women's reproductive healthcare.



1 Innovation & Differentiation

Home-Based, Wearable Ultrasound
Enables regular uterine imaging outside hospital settings.

First dedicated wearable solution for polyp monitoring

AI-Enabled Analysis
Automated assessment of polyp size and morphology from ultrasound images.

Reduces dependence on frequent expert interpretation

3D Digital Twin
Creates a patient-specific 3D model for better visualisation and comparison over time.

Provides longitudinal and personalised insights

Patient-Centric Design
Designed to be user-friendly, comfortable and suitable for repeated home use.

Improves accessibility, convenience and continued monitoring

2 Future Potential

Clinical Validation & Adoption
Multi-centre studies to validate accuracy, reliability and real-world usability.

Pathway to clinical use

Expanded Indications
Potential use in other gynaecological conditions such as fibroids, cysts and adenomyosis.

Broader women's health monitoring platform

Integration with Care Ecosystem
Link with telemedicine platforms, EHR systems and clinical decision-support tools.

End-to-end digital care pathway

Scalable Impact
Potential to enable larger population screening and long-term reproductive health monitoring globally.

Towards accessible, data-driven women's healthcare

POLYPULSE

Thank You

For your time, support and belief in
a safer, healthier future for women.

POLYPULSE | EARLIER DETECTION
CLEARER INSIGHTS
HEALTHIER TOMORROWS

Team Eternia

- Priyanshi Rathore
- Yash Kumar Khatik
- Ishan Khan
- Gaurav Shah

TOGETHER FOR A HEALTHIER TOMORROW FOR EVERY WOMAN